United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam Fall 2025

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: **40** Duration: 2 Hours

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. Consider the following Context-free grammars (CFG) and answer according to it: 5x2

a) With the help of **Top-Down Parse Tree**, derive the following string: (aa(aba)bb)(b(abb)b)

$S \rightarrow (S) \mid SS \mid A \mid a \mid b$
$A \rightarrow aAa \mid bAb \mid S \mid aAb \mid bAa$

b) With the help of **rightmost derivations**, find out if the grammar is **ambiguous or not** for the string: { r % r } - r % r

$X \rightarrow X \% Y \mid Y$
$Y \rightarrow Y - Z \mid Z$
$Z \rightarrow r \mid \{X\} \mid X$

2.5x4

2. Design CFGs that generate the following languages:

a) $L = \{ a^p b^q c^r \mid p+q = 2r \text{ and } p, q, r \geq 1 \}$

b) $L = \{ 0^x 1^{2y} 2^z \mid x = z - 2 \text{ and } x, z \geq 1, y > 1 \}$

c) L = { w ∈ { x, y, z }* | w contains **xzy** as a substring and the third last symbol of w is **x** }

d) L = { w ∈ { a, b }* | every even length palindrome when the string starts with **a** or every odd length palindrome when the string starts with **b** }

3 Draw the Push Down Automata (PDA) for the following languages: 5x2

a) $L = \{ 0^x \# 1^y \mid \Sigma = \{ 0, 1, \# \} ; x \geq 3y ; x,y \geq 1 \}$

b) $L = \{ a^p b^q c^r d^s \mid \Sigma = \{ a, b, c, d \} ; p+q = 2(r+s) ; p,q,r,s \geq 0 \}$

4. Draw a Turing machine for the following languages : 5x2

a) $L = \{ a^{2n} b^m c^{3n} d^{m+2} \mid n \geq 1, m \geq 2 \}$

b) $L = \{ w \# w \mid w \in \{ a, b, c \}^+ \}$